

PROJECT DEFENCE

Model Questions & Answers

Zambian Green Energy Sentiment Analysis Prototype

40 MODEL QUESTIONS

COMPLETED PROTOTYPE

Prepared for viva and project demonstration

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Thirty-Second Opening

This project developed a locally adapted sentiment-analysis prototype for public Facebook comments about ZESCO and electricity-service experiences in Zambia. It collects public top-level comments, removes unnecessary personal references, filters comments for energy relevance, detects sentiment, sarcasm, topics, and emotions, and presents explainable findings through a mobile-responsive Streamlit dashboard. Its main contribution is adapting standard social-media sentiment methods to Zambian and code-switched expressions while preserving an auditable link between dashboard findings and source comments.

Core Project Questions

1. What problem does the project solve?

MODEL ANSWER

It helps analysts summarize large volumes of public electricity-service feedback that would be slow and inconsistent to review manually. The system highlights sentiment, emotions, topics, and changes across discussions, while keeping the underlying comments available as evidence.

2. What is the main objective?

MODEL ANSWER

The main objective is to design and evaluate an explainable sentiment-analysis prototype for Zambian electricity discourse that can process public Facebook comments and communicate policy-relevant findings through an accessible dashboard.

3. Why focus on ZESCO and electricity comments?

MODEL ANSWER

Electricity reliability, tariffs, faults, and load shedding directly affect households and businesses. These topics generate substantial online discussion, making them a relevant case for testing whether locally adapted sentiment analysis can support public-service monitoring.

4. Why use Facebook comments?

MODEL ANSWER

Facebook contains unsolicited and timely public reactions to announcements. It is useful for studying expressed concerns, but the project does not claim that Facebook users represent the whole Zambian population.

5. What is the end-to-end system flow?

MODEL ANSWER

A public Facebook URL is submitted, Apify retrieves top-level comments, the system cleans and anonymises the text, filters for energy relevance, applies sentiment, sarcasm, emotion, topic, and machine-learning analysis, saves the result, and presents aggregated evidence through the dashboard and downloadable reports.

6. What makes the project locally relevant?

MODEL ANSWER

It adds Zambian and code-switched sentiment and emotion lexicons, preserves readable context for negation, and introduces local sarcasm patterns and energy-specific relevance terms. This reduces errors caused by applying a generic English tool without adaptation.

7. What is the main contribution or novelty?

MODEL ANSWER

The contribution is the integration of locally adapted lexical rules, explainable relevance and topic taxonomies, sarcasm-aware corrections, classical models, and evidence-linked visual reporting in one working Zambian energy-discourse prototype.

Data and Ethics Questions

8. How was the data collected?

MODEL ANSWER

The system uses Apify to collect publicly accessible, top-level comments from supplied Facebook post URLs. Each request is capped at 1,000 comments, and cached comments are reused to reduce repeated collection and token consumption.

9. Why exclude replies?

MODEL ANSWER

Replies often contain side conversations, repeated names, and context that is not directly about the original post. Restricting the study to top-level comments gives a clearer and more consistent unit of analysis.

10. How does the system address privacy?

MODEL ANSWER

It analyzes public comments, does not intentionally store commenter profile names, removes structured mentions and @ mentions, reports aggregated findings, and exposes comments only as supporting evidence. It should still be treated as a research prototype governed by the institution's ethics requirements.

11. Is the dataset representative of Zambia?

MODEL ANSWER

No. It represents people who chose to comment on selected public Facebook posts. The findings describe observed online discourse and must not be presented as population-level opinion or as a substitute for a representative survey.

12. How are irrelevant comments handled?

MODEL ANSWER

An explainable keyword-based relevance stage identifies electricity, ZESCO, load-shedding, tariff, infrastructure, and related terms. Irrelevant comments are retained for audit but are excluded from model training and the default policy-facing analysis.

Preprocessing and Feature Questions

13. What preprocessing is performed?

MODEL ANSWER

The pipeline removes URLs and mentions, handles emojis, normalizes text, tokenizes, removes stop words where appropriate, lemmatizes tokens, and preserves readable text for rules that depend on negation, phrases, or sarcasm.

14. Why preserve negation and readable text?

MODEL ANSWER

Removing words such as "not" can reverse meaning. For example, "power is not stable" must not be reduced to "power stable." The system therefore keeps a context-preserving text form for sentiment rules while also generating processed text for TF-IDF models.

15. How are code-switched expressions handled?

MODEL ANSWER

Local lexicon CSV files map Zambian or code-switched words and phrases to sentiment and emotion weights. Matches are recorded in explanatory columns so a reviewer can see why a local correction was made.

16. How is sarcasm detected?

MODEL ANSWER

The prototype uses transparent heuristic patterns, including gratitude followed by a complaint, positive wording combined with laughter, mocked promises, rhetorical questions, and selected local intensifiers. Sarcasm can change apparently positive language to negative and can suppress false hope or trust scores.

Sentiment and Emotion Questions

17. Why use VADER?

MODEL ANSWER

VADER is designed for short, informal social-media text and provides an interpretable compound sentiment score. It is lightweight enough for an interactive prototype, and its limitations can be addressed through explicit local correction rules.

18. What labels does the system produce?

MODEL ANSWER

Sentiment is classified as positive, neutral, or negative. The system also estimates anger, fear, trust, sadness, hope, and frustration, and assigns one explainable energy-policy topic to each relevant comment.

19. Why include emotions in addition to sentiment?

MODEL ANSWER

Polarity alone cannot distinguish anger from fear or frustration from sadness. Emotion scores provide more useful context for communication and service-response decisions.

20. What topics are detected?

MODEL ANSWER

The taxonomy includes load shedding and reliability, tariffs and affordability, renewable energy and solar, customer service and communication, faults and infrastructure, governance and trust, jobs and economic impact, environment and climate, and other energy concerns.

Machine-Learning Model Questions

21. Which machine-learning models were trained?

MODEL ANSWER

The project trains Multinomial Naive Bayes, Logistic Regression, and a linear Support Vector Machine. They are appropriate, well-understood baselines for sparse TF-IDF text features.

22. What is TF-IDF and why was it used?

MODEL ANSWER

TF-IDF converts text into numerical features by increasing the importance of terms that are frequent in a document but less common across the corpus. It is efficient, interpretable, and works well with classical text classifiers.

23. How was data leakage prevented?

MODEL ANSWER

Training and testing are separate persisted files. The vectorizer and models are fitted only on the 80% training split, while metrics and confusion matrices are generated using the held-out 20% test split. The split uses a fixed random state for reproducibility.

24. What is the class distribution?

MODEL ANSWER

The relevant VADER-labeled dataset contains 2,813 comments: 1,011 negative, 904 neutral, and 898 positive. The classes are not perfectly equal, so weighted precision, recall, and F1 are reported alongside accuracy.

25. What were the VADER-branch model results?

MODEL ANSWER

On the held-out relevant-comment test split, Naive Bayes achieved 55.06% accuracy and 53.70% weighted F1; Logistic Regression achieved 58.26% accuracy and 58.03% weighted F1; and SVM achieved 58.44% accuracy and 58.29% weighted F1. SVM was marginally strongest on both measures.

26. What were the optional RoBERTa-labeled branch results?

MODEL ANSWER

The alternate branch produced 66.61% accuracy for Naive Bayes, 72.29% for Logistic Regression, and 71.23% for SVM. Its weighted F1 scores were 56.32%, 69.59%, and 69.89%, respectively. These results come from a different labeling branch and should not be presented as a direct, controlled replacement for the corrected VADER branch without fresh common-label evaluation.

27. Which model is best?

MODEL ANSWER

For the corrected VADER-labeled branch used by the main workflow, linear SVM is the strongest baseline by a small margin. In the optional RoBERTa-labeled branch, Logistic Regression has the highest accuracy while SVM has the highest weighted F1. "Best" therefore depends on the labeling branch and evaluation measure.

28. Why is accuracy alone insufficient?

MODEL ANSWER

Accuracy can hide poor performance on a smaller class. Weighted precision, recall, F1, the classification report, and confusion matrices show whether a model performs consistently across positive, neutral, and negative classes.

29. Why are the baseline scores not higher?

MODEL ANSWER

The task contains code-switching, spelling variation, sarcasm, short ambiguous comments, and automatically assisted labels rather than a large expert-annotated gold standard. The scores are honest baseline results and motivate better annotation and contextual modeling.

30. Why not deploy only RoBERTa?

MODEL ANSWER

The prototype prioritizes transparency, modest computing requirements, and reproducibility. The RoBERTa branch is retained as an experiment, but the locally corrected VADER workflow and classical models are easier to explain and run during a live defence. A future controlled comparison on one human-reviewed test set would support a stronger deployment decision.

Dashboard and Trend Questions

31. What does the dashboard provide?

MODEL ANSWER

It provides URL analysis, a sentiment and emotion overview, trend charts, explainable topics, comment-level evidence, and downloadable stakeholder reports. It is designed for analysts and communicators rather than model developers.

32. How does five-link trend tracking work?

MODEL ANSWER

The Trends tab loads up to five distinct saved links, orders them by Facebook metadata dates, and compares sentiment percentages across those discussions. It uses the post-publication timestamp when available and otherwise the earliest valid Facebook comment timestamp. It does not call Apify and never substitutes the date on which the app performed the analysis.

33. Can the system claim that an event caused sentiment to change?

MODEL ANSWER

No. A post title, topic, or timing can help explain what was being discussed, but the chart shows association rather than causation. External operational evidence would be required to attribute a change to a tariff decision, outage, or policy event.

34. How is the mobile interface supported?

MODEL ANSWER

At phone widths, multi-column sections stack vertically, tab navigation becomes horizontally scrollable, touch controls use larger targets, headings and page padding shrink, and wide tables remain scrollable. The analytical content remains the same as on desktop.

35. What happens when Apify fails?

MODEL ANSWER

Technical details are logged on the server, while the user sees a short, safe message. Apify failures are presented as token depletion as requested, and cached analyses remain available without making another Apify call.

Evaluation, Limitations, and Future Work

36. How was the software validated?

MODEL ANSWER

Automated tests cover collection limits, reply exclusion, caching, preprocessing, relevance, local sentiment, sarcasm, emotions, model utilities, reports, saved-link trends, metadata-date selection, and user-friendly errors. The final interface is also checked at desktop and mobile viewports.

37. What are the main limitations?

MODEL ANSWER

The main limitations are selected-post sampling, automatically assisted labels, heuristic sarcasm and relevance rules, incomplete coverage of local language, dependence on public Facebook accessibility and Apify, and ephemeral file storage on a free Render deployment.

38. What is the most important future improvement?

MODEL ANSWER

Create a human-annotated validation set of 200-500 diverse comments with agreement between multiple reviewers. That would provide a stronger gold standard for relevance, sentiment, sarcasm, and emotion evaluation.

39. How should production persistence be improved?

MODEL ANSWER

Move link history and analyzed comments from local CSV files to persistent storage such as a managed PostgreSQL database or a paid persistent disk. This is necessary for dependable trend tracking across restarts and month-long monitoring periods.

40. What is the final conclusion?

MODEL ANSWER

The project demonstrates that a locally adapted and explainable pipeline can convert noisy public electricity discourse into structured sentiment, emotion, topic, and trend evidence.

It is suitable as a completed research prototype, while its documented limitations define a clear path toward stronger validation and production deployment.

Short Demonstration Sequence

1. Paste a public Facebook post URL and select the comment limit.
2. Explain the cache check and top-level-comment restriction.
3. Show sentiment, dominant emotion, and leading topic in Overview.
4. Show the current discussion pattern and five-link comparison in Trends.
5. Open Topics and Comments to demonstrate explainability.
6. Download the stakeholder report.
7. End by stating the sampling, causality, annotation, and persistence limitations.

Answers to Avoid



- Do not say that Facebook comments represent all Zambians.
- Do not claim that the system proves an external event caused a sentiment change.
- Do not describe automatically assisted labels as a human-annotated gold standard.
- Do not compare VADER-branch and RoBERTa-branch scores as if only the classifier changed.
- Do not promise that free Render storage preserves month-to-month history.
- Do not describe the dashboard's policy considerations as automatic policy decisions.